

LAPORAN PRAKTIKUM – BUBBLE SORT

Laporan ini disusun untuk memenuhi Tugas Ke Tujuh Praktikum Algoritma & Pemrograman

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PROGRAM STUDI TEKNIK INFORMATIKA

FAKULTAS TEKNIK

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Soal Bubble Sort_06 (nama file: PASD7-06.cpp):

- **Source Code**

6. Berdasarkan contoh no1, urutkan secara Ascending dengan tanpa inputan (nama file : PASD7-06.cpp):

```
#include <iostream>
using namespace std;

int data[10], data2[10];
int n;

void tukar(int a, int b){
    int t;
    t = data[b];
    data[b] = data[a];
    data[a] = t;
}

void input(){

}

void tampil(){
    for(int i=0; i<n; i++){
        cout<<data[i]<<" ";
    }
    cout<<endl;
}

void BubbleSort(){
    for(int i=1; i<n; i++){
        for(int j=n-1; j>=i; j--){
            if(data[j] < data[j-1]){
                tukar(j, j-1);
            }
        }
    }
}
```

```

        }
        tampil();
    }
    cout<<endl;
}
}

int main(){

    cout<<"*-----*"<<endl;
    cout<<"Bubble Sort Ascending"<<endl;
    cout<<"*-----*"<<endl;
    n=10;
    int temp[10]={45,25,21,0,0,35,11,6,5,21};
    for(int i=0;i<n;i++){
        data[i]=temp[i];
        data2[i]=data[i];
    }
    cout<<"Data Awal : ";
    tampil();
    cout<<"\nProses Bubble Sort"<<endl;
    BubbleSort();
    cout<<"\nHasil Akhir : ";
    tampil();
    return 0;
}

```

- SS Program

```
#include <iostream>
using namespace std;

int data[10], data2[10];
int n;

void tukar(int a, int b){
    int t;
    t = data[b];
    data[b] = data[a];
    data[a] = t;
}

void input(){

}

void tampil(){
    for(int i=0; i<n; i++){
        cout<<data[i]<<" ";
    }
    cout<<endl;
}

void BubbleSort(){
    for(int i=1; i<n; i++){
        for(int j=n-1; j>=i; j--){
            if(data[j] < data[j-1]){
                tukar(j, j-1);
            }
        }
        tampil();
    }
    cout<<endl;
}

int main(){

    cout<<"*-----*"<<endl;
    cout<<"Bubble Sort Ascending"<<endl;
    cout<<"*-----*"<<endl;
    n=10;
    int temp[10]={45,25,21,0,0,35,11,6,5,21};
    for(int i=0;i<n;i++){
        data[i]=temp[i];
        data2[i]=data[i];
    }
    cout<<"Data Awal : ";
    tampil();
    cout<<"\nProses Bubble Sort"<<endl;
    BubbleSort();
    cout<<"\nHasil Akhir : ";
    tampil();
    return 0;
}
```

- SS Running Program

```
D:\ASD SMT2\ASD-A-25-26\PRAK7_Bubble Sort_Asprak01_4525210035_Khairul Adam Efendi>PASD7-06
*-----*
Bubble Sort Ascending
*-----*
Data Awal : 45 25 21 0 0 35 11 6 5 21

Proses Bubble Sort
45 25 21 0 0 35 11 6 5 21
45 25 21 0 0 35 11 5 6 21
45 25 21 0 0 35 5 11 6 21
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0 0 5 6 11 21 45 21 25 35
0 0 5 6 11 21 21 45 25 35

0 0 5 6 11 21 21 25 45 35
0 0 5 6 11 21 21 25 45 35

0 0 5 6 11 21 21 25 35 45

Hasil Akhir : 0 0 5 6 11 21 21 25 35 45
```

Soal Bubble Sort_07 (nama file: PASD7-07.cpp):

- **Source Code**

7. Berdasarkan contoh no1, urutkan secara Descending tanpa inputan (nama file : PASD7-07.cpp):

```
#include <iostream>
using namespace std;

int data[10], data2[10];
int n;

void tukar(int a, int b){
    int t;
    t = data[b];
    data[b] = data[a];
    data[a] = t;
}

void input(){

}

void tampil(){
    for(int i=0; i<n; i++){
        cout<<data[i]<<" ";
    }
    cout<<endl;
}

void BubbleSort(){
    for(int i=1; i<n; i++){
        for(int j=n-1; j>=i; j--){
            if(data[j] > data[j-1]){
                tukar(j, j-1);
            }
        }
    }
}
```

```

        }
        tampil();
    }
    cout<<endl;
}
}

int main(){

    cout<<"*-----*"<<endl;
    cout<<"Bubble Sort Ascending"<<endl;
    cout<<"*-----*"<<endl;
    n=10;
    int temp[10]={45,25,21,0,0,35,11,6,5,21};
    for(int i=0;i<n;i++){
        data[i]=temp[i];
        data2[i]=data[i];
    }
    cout<<"Data Awal : ";
    tampil();
    cout<<"\nProses Bubble Sort"<<endl;
    BubbleSort();
    cout<<"\nHasil Akhir : ";
    tampil();
    return 0;
}

```

- SS Program

```
#include <iostream>
using namespace std;

int data[10], data2[10];
int n;

void tukar(int a, int b){
    int t;
    t = data[b];
    data[b] = data[a];
    data[a] = t;
}

void input(){

}

void tampil(){
    for(int i=0; i<n; i++){
        cout<<data[i]<<" ";
    }
    cout<<endl;
}

void BubbleSort(){
    for(int i=1; i<n; i++){
        for(int j=n-1; j>=i; j--){
            if(data[j] > data[j-1]){
```



```

        tukar(j, j-1);
    }
    tampil();
}
cout<<endl;
}
}

int main(){

    cout<<"*-----*"<<endl;
    cout<<"Bubble Sort Ascending"<<endl;
    cout<<"*-----*"<<endl;
    n=10;
    int temp[10]={45,25,21,0,0,35,11,6,5,21};
    for(int i=0;i<n;i++){
        data[i]=temp[i];
        data2[i]=data[i];
    }
    cout<<"Data Awal : ";
    tampil();
    cout<<"\nProses Bubble Sort"<<endl;
    BubbleSort();
    cout<<"\nHasil Akhir : ";
    tampil();
    return 0;
}

```

- SS Running Program

```
D:\ASD SMT2\ASD-A-25-26\PRAK7_Bubble Sort_Asprak01_4525210035_Khairul Adam Efendi>PASD7-07
*-----*
Bubble Sort Ascending
*-----*
Data Awal : 45 25 21 0 0 35 11 6 5 21

Proses Bubble Sort
45 25 21 0 0 35 11 6 21 5
45 25 21 0 0 35 11 21 6 5
45 25 21 0 0 35 21 11 6 5
45 25 21 0 0 35 21 11 6 5
45 25 21 0 35 0 21 11 6 5
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45 35 25 21 21 11 6 5 0 0

Hasil Akhir : 45 35 25 21 21 11 6 5 0 0
```